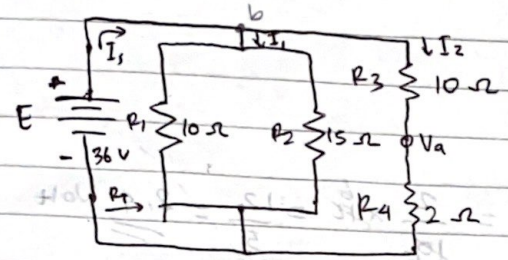
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NIM : 221331002
Kelas : 1A - TTE
Mata kuliah : Rangkaian Listrik 1

10 Oktober 2022

1.A. Tentukan besarnya nilai I_s, I_1, I_2, V_a dan R_T.



$R_1 // R_2 = \frac{10 \cdot 15}{10+15} = \frac{150}{25} = 6\ \Omega$

- node b

$36 = \frac{V_b}{6} + \frac{V_b - V_a}{10} \quad \Rightarrow \times 30$

$1080 = 5V_b + 3V_b - 3V_a$

$1080 = -3V_a + 8V_b \ldots ①$

- node a

$\frac{V_b - V_a}{10} = \frac{V_a}{2} \quad \Rightarrow \times 10$

$V_b - V_a = 5V_a$

$V_b = 6V_a \ldots ②$

- subs ② ke ①

$1080 = -3V_a + 8(6V_a)$

$1080 = -3V_a + 48V_a$

$1080 = 45V_a$

$V_a = 24$ Volt

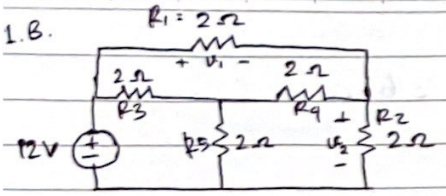
- $V_b = 6(24)$

$= 144$ Volt

- $I_1 = \frac{V_b}{6} = \frac{144}{6} = 24$ A

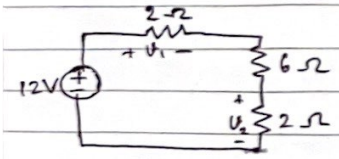
- $I_2 = \frac{V_b - V_a}{10} = \frac{144 - 24}{10} = \frac{120}{10} = 12$ A

1.B.



$R_{35seri} = 2+2 = 4\Omega$

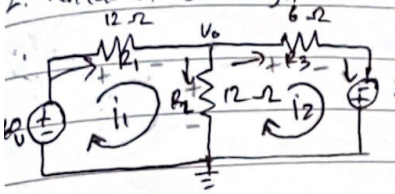
$R_{35seri}\,R_4 = 4+2 = 6\Omega$



$$V_1 = \frac{R_1}{R_1+R_2+R_3} \times V_S = \frac{2}{2+2+6} \times 12 = \frac{2}{10} \times 12 = \frac{12}{5} = 2,4 \text{ Volt}$$

$$V_2 = \frac{R_2}{R_1+R_2+R_3} \times V_S = \frac{2}{2+2+6} \times 12 = \frac{2}{10} \times 12 = \frac{12}{5} = 2,4 \text{ Volt}$$

2. Tentukan besarnya V_o dan daya di masing-masing Resistor.



$V_1 = 12 i_1 \Rightarrow V_1 = 12 \times 4,75 = 57$ Volt

$V_2 = 12 (i_1 - i_2) \Rightarrow V_2 = 12 (4,75 - 4.5) = 12(0,25) = 3$ Volt

$V_3 = 6 i_2 \Rightarrow V_3 = 6 \times 4,5 = 27$ Volt

• loop i_1

$-60 + V_1 + V_2 = 0$

$-60 + 12i_1 + 12i_1 - 12i_2 = 0$

$24i_1 - 12i_2 = 60 \dots$ ①

• loop i_2

$-24 - V_2 + V_3 = 0$

$-24 - 12i_1 + 12i_2 + 6i_2 = 0$

$-12i_1 + 18i_2 = 24 \dots$ ②

- elim i_1

$24i_1 - 12i_2 = 60$

$-12i_1 + 18i_2 = 24 \quad | \times 2 |$

$24i_1 - 12i_2 = 60$

$-24i_1 + 36i_2 = 48$ +

$24i_2 = 108$

$i_2 = 4,5$ A ✓

• subs ke ①

$24i_1 - 12(4,5) = 60$

$24i_1 - 54 = 60$

$24i_1 = 114$

$i_1 = 4,75$ A ✓

• $V_o = 57 - 3 - 27 = 27$ Volt ✓

• Daya

$P_{R1} = 57 \times 4,75 = 270,75$ watt ✓

$P_{R2} = \frac{3^2}{12} = \frac{9}{12} = \frac{3}{4} = 0,75$ watt ✓

$P_{R3} = 27 \times 4,5 = 121,5$ watt. ✓